

PORTFOLIO

25 ▸ 26

Industrial Design

Hello Everybody!

Here are a few things you should know about me:

Ketaki Khadilkar, 20

I’m a Third year Product Design student at MIT-WPU. I love interacting with new people and my current interests include studying products which includes complex mechanisms and finding innovative solutions to improve product functionality and improvise on ease of use.

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EDUCATION

2009 -2021
SPM English School, Pune

2021 -2023 (HSC, PCM.)
S.P college, Pune

2023-2027
(B.Des, PD/ID) MIT-WPU

EXPERIENCE

- Vegapod Hyperloop: Technical & Media Team
- Project Lead: Echo Bloom, Harit Setu
- Cosmos Club: Design & Media Team
- Volunteer: DevConf 2025 (Red Hat)
- E Summit Design Team

INTERESTS

- Playing Sports
- Crafting and Painting
- Graphic Design
- Stitching and Knitting
- Playing LEGO
- Watching F1

SKILLS

PRODUCT

- Solidworks
- Fusion
- AutoCad
- Rhino
- Keyshot

GRAPHIC

- Figma
- Illustrator
- Photoshop
- Lightroom
- Canva

SOFT

- Technical Design
- Material Knowledge
- Problem Solving
- Concept Visualisation
- Prototyping and Sculpting

CERTIFICATES

- Teamwork and Collaboration
- Measuring Sustainable Development
- Critical Thinking and Problem Solving

SLIQUE

Project Duration: 4 months
Type: Product design studio project
Team: Individual Project

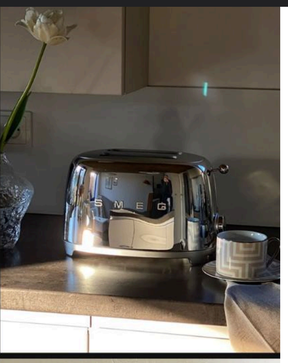


Design for All

Efficiency outweighs the morning crunch?
(High-pressure reality of the start of the day.)

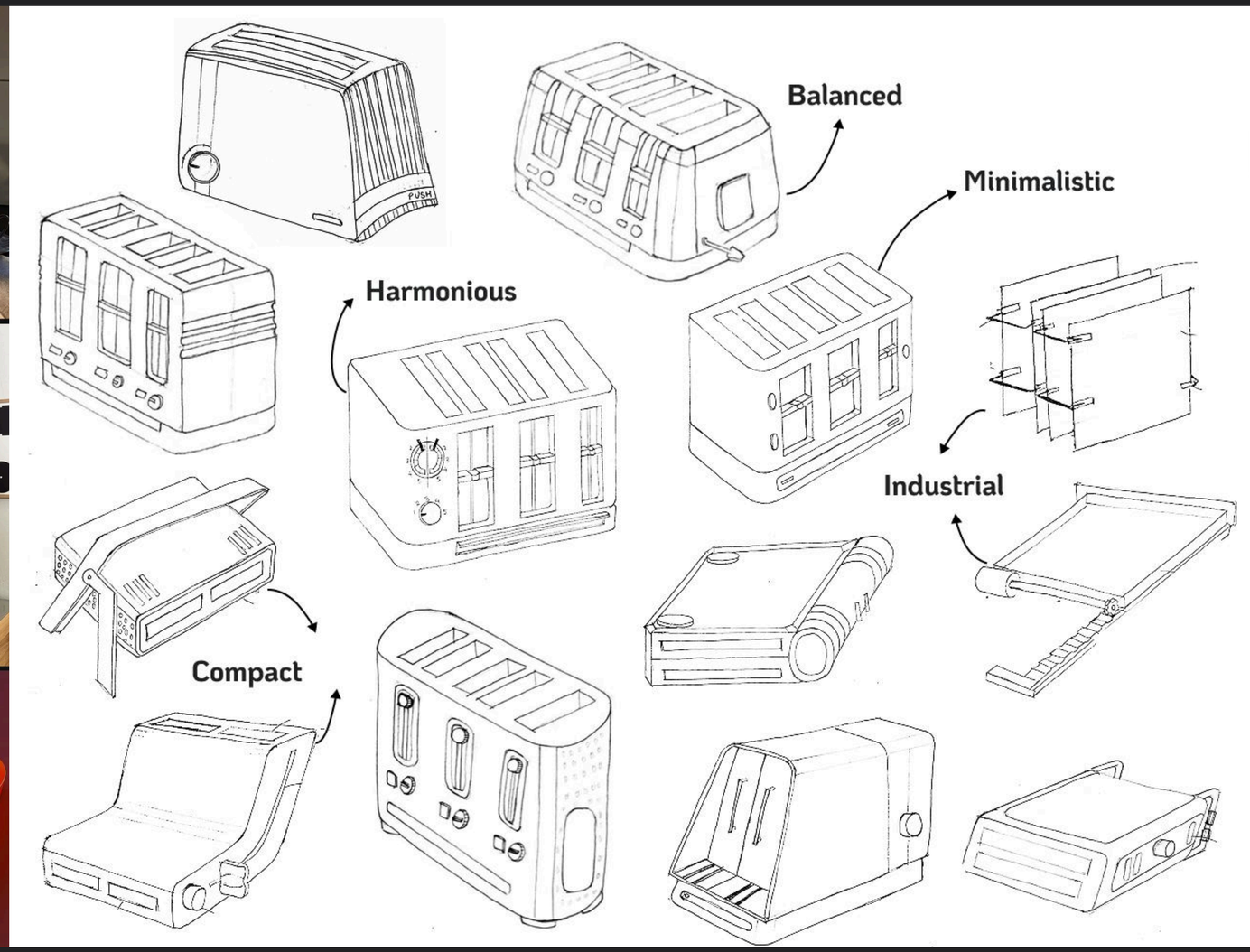


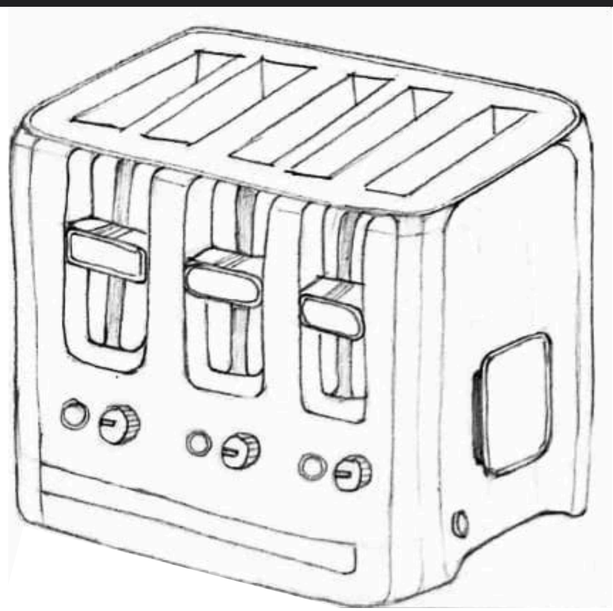
Ideation







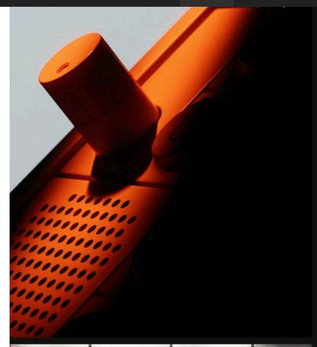


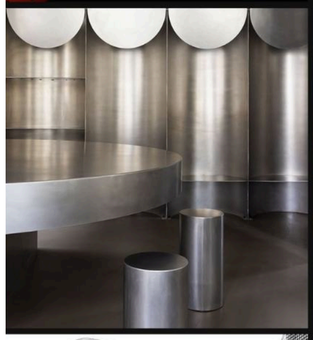


Consistent browning components with compact body and sturdy built

Easier cleaning with visual indication components

Multiple toasting slots with individual functioning

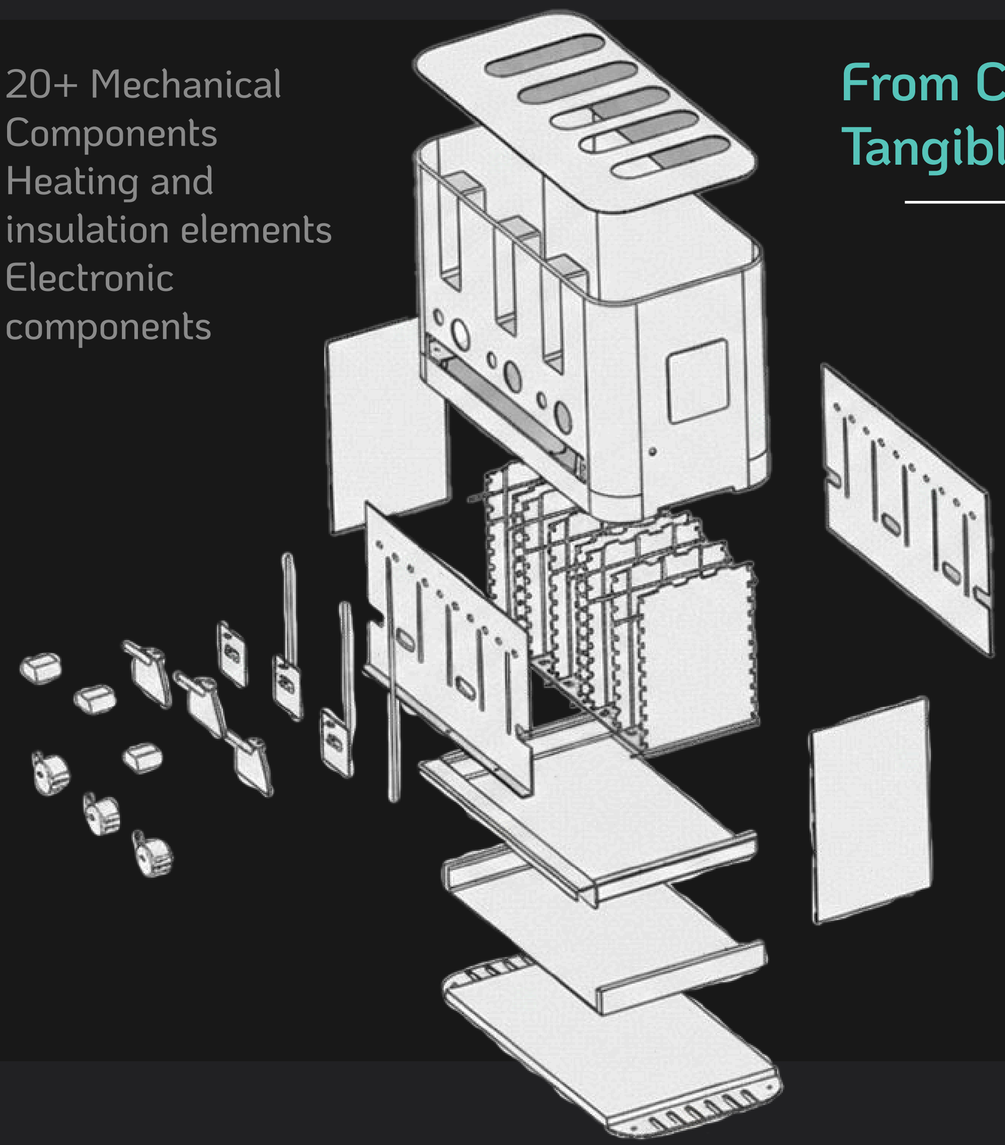






How it’s made?

- 20+ Mechanical Components
- Heating and insulation elements
- Electronic components

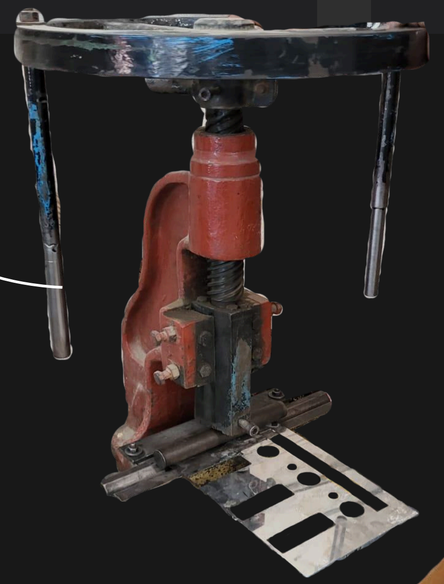


From CAD to Tangible Product

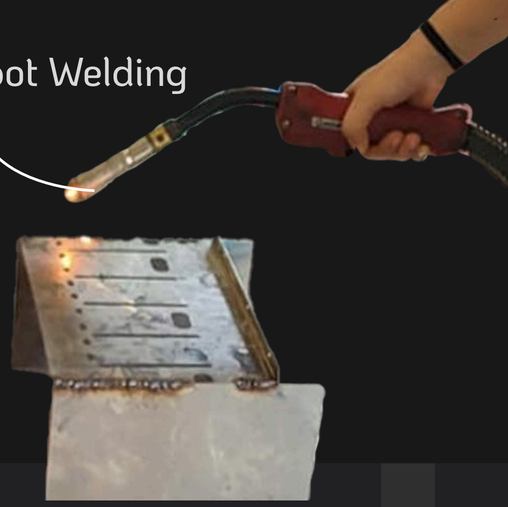


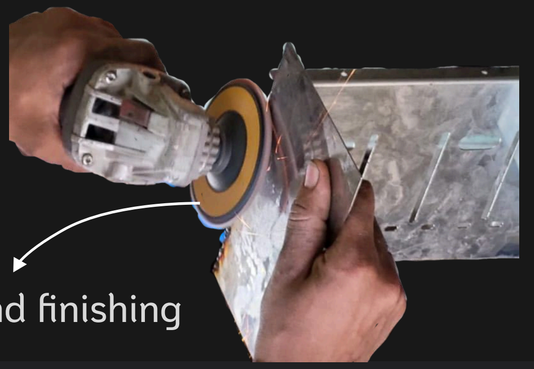
Laser cutting

Hand press bending



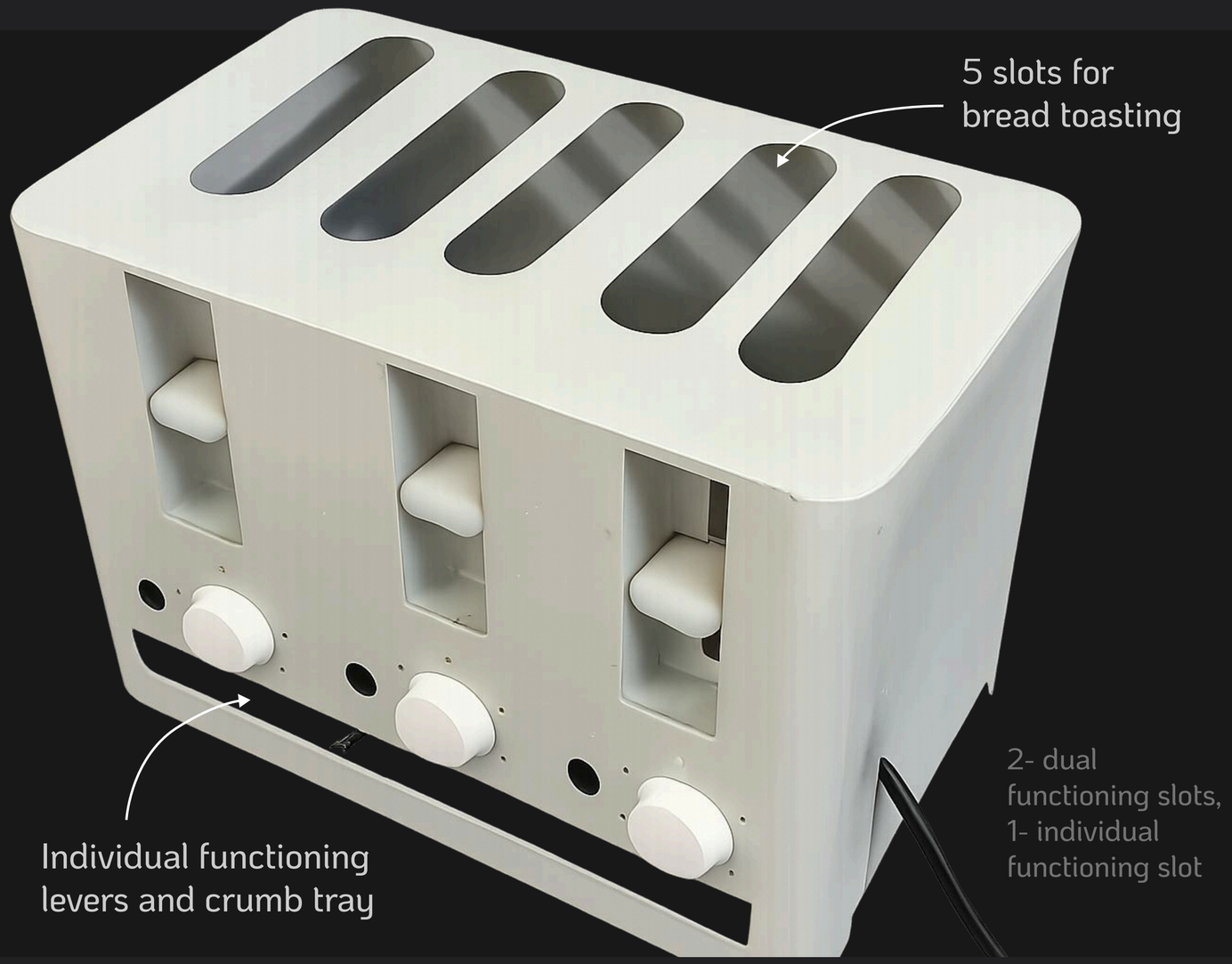
CO2 Spot Welding





Grinding and finishing

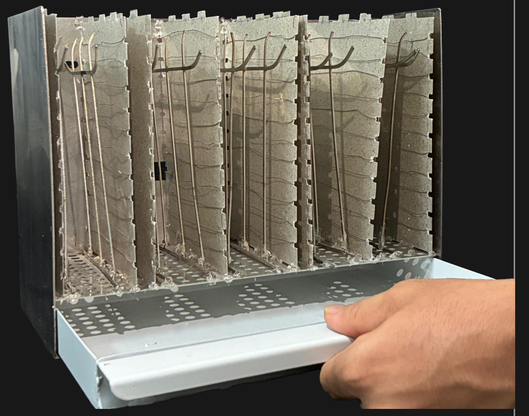
Prototyping



Component List:

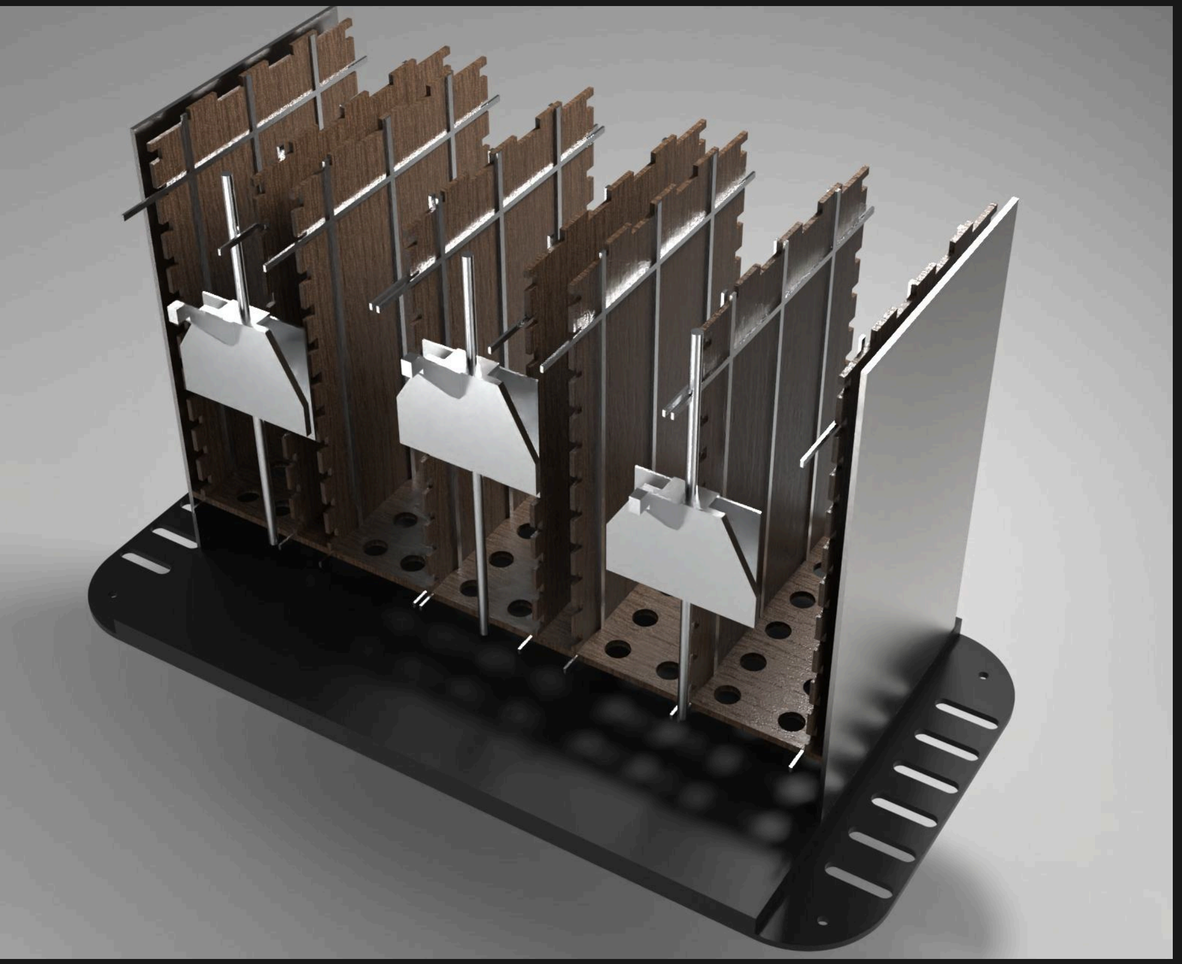
- Main body: Stainless steel
- Lever: ABS plastic
- Control Knob: Plastic
- Crumb tray: Stainless steel

- Inner body: Stainless steel
- Lever control rod: Stainless steel
- Spring:
- Heating element: Nichrome wire
- Insulation material: Mica sheet



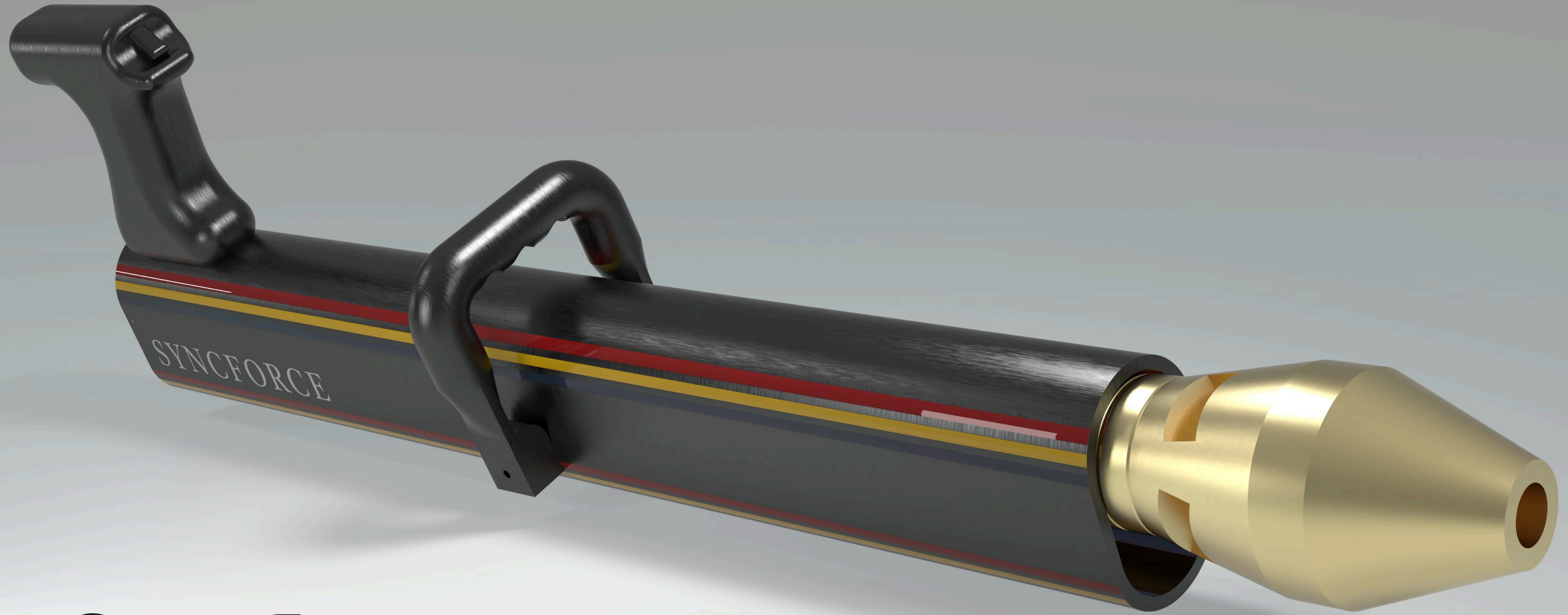


Product Detailing






SLOT 1
SLOT 2
SLOT 3
CANCEL



SyncForce

Project Duration: 4 months
Type: Product design studio project
Team: Individual Project

For The Society

Precision in limited time span
(High-pressure reality of Fire Fighters and Rescue Team.)

“For calls in any other branch, there is no special equipment to carry out faster rescue in car accidents”

“Equipment provided to firefighters are simple and often have a very niche, and singular role. Because of this, the amount of equipment needed is very large”



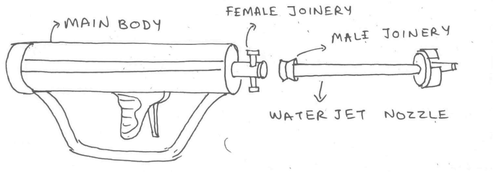
“Firetrucks and rescue trucks are highly equipped with specialized tools, but this variety leads to in-organization and chaos within the truck.”

Ideation

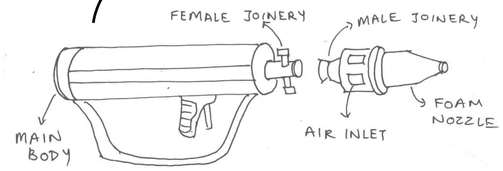


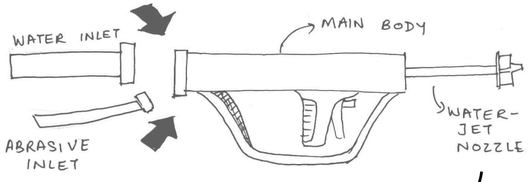




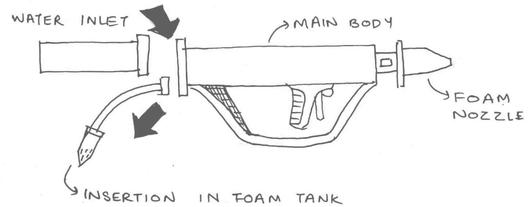


Effective Response

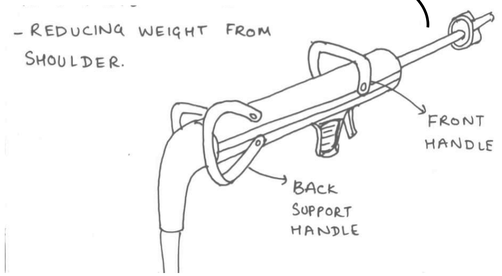


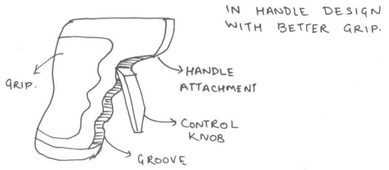


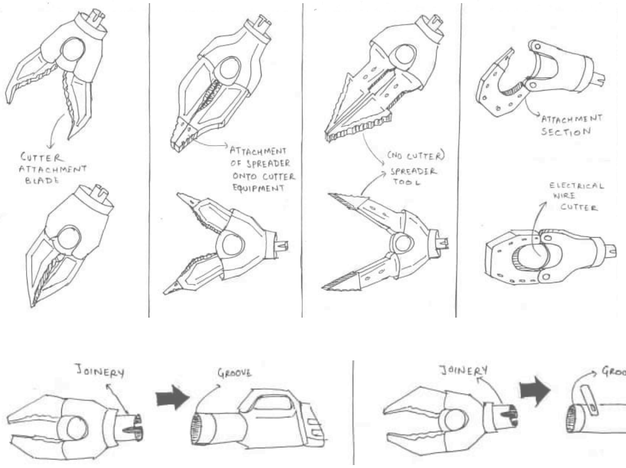
Faster operation

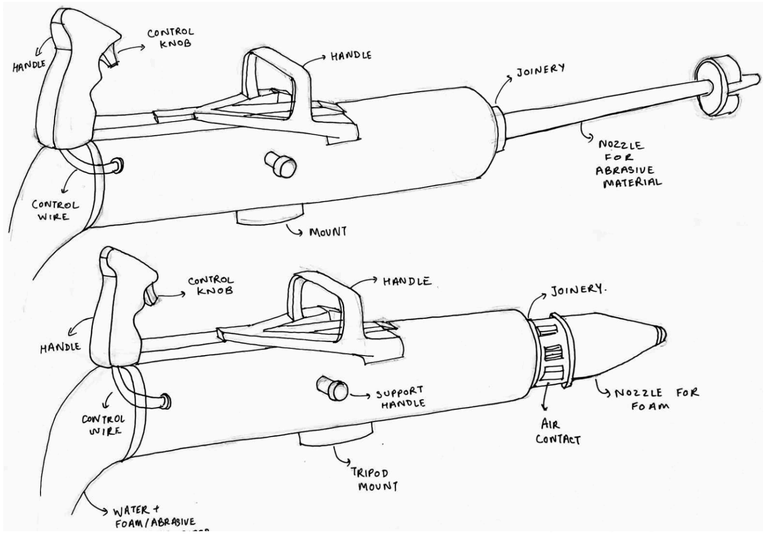


Reduced Fatigue

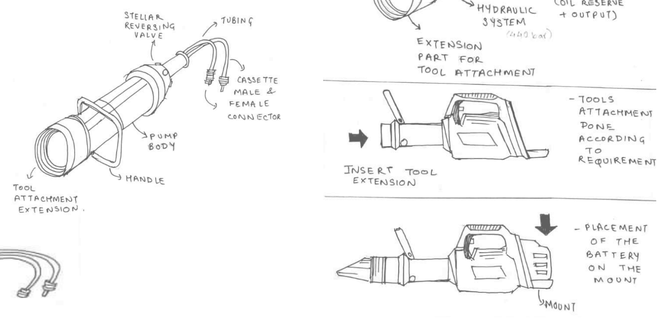


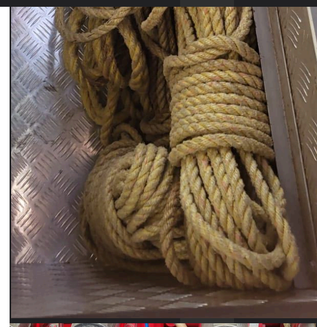


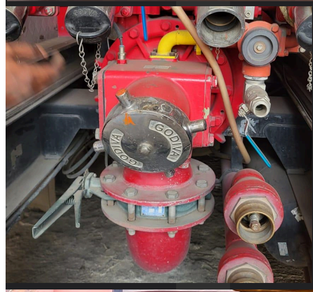




Modular











Market Research





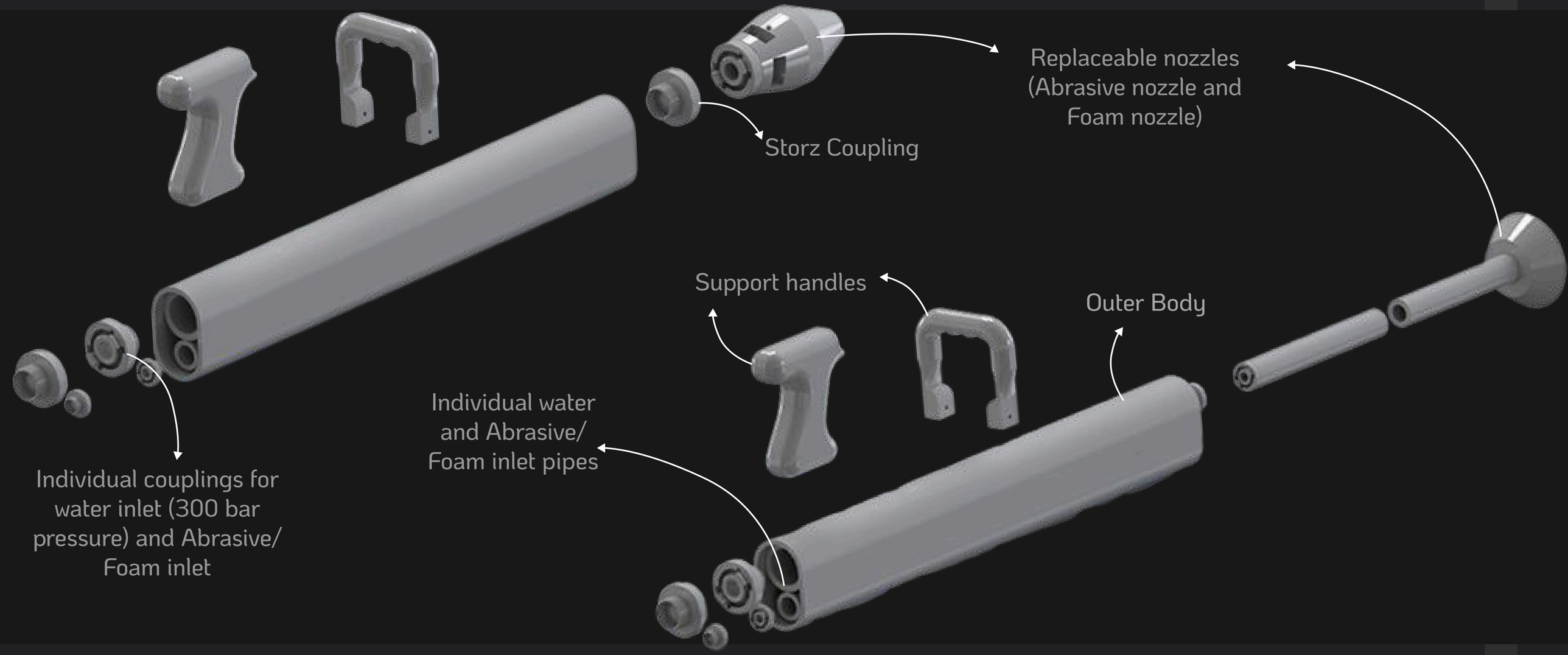




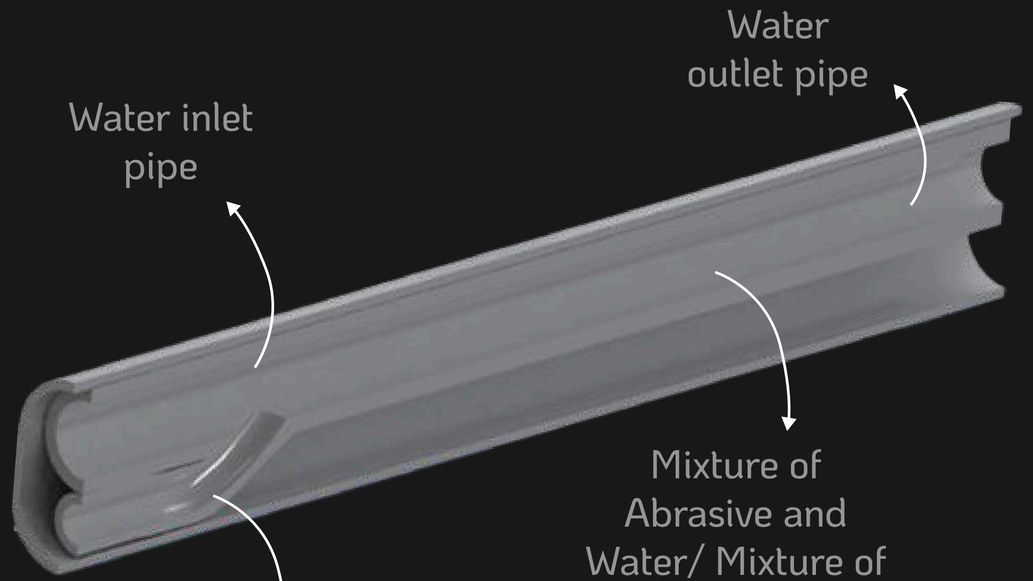
C 360 H Kit

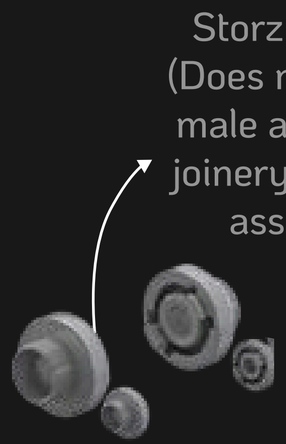
- The C 360 H Kit is a **cutting extinguisher system** specifically engineered for vehicle-based installation in larger fire engines. The kit includes all essential components up to the hand lance.
- **Uses truck engine as a Power Source**
- Requires hydraulic Power and Water Supply from vehicle

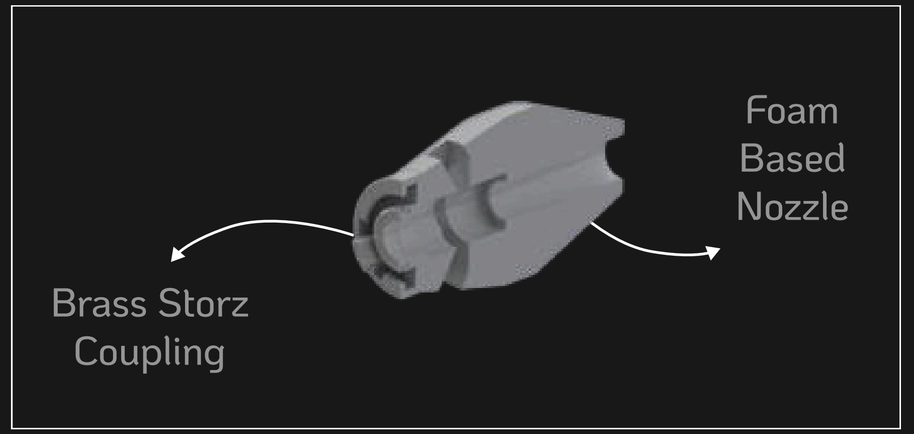
Combination tool

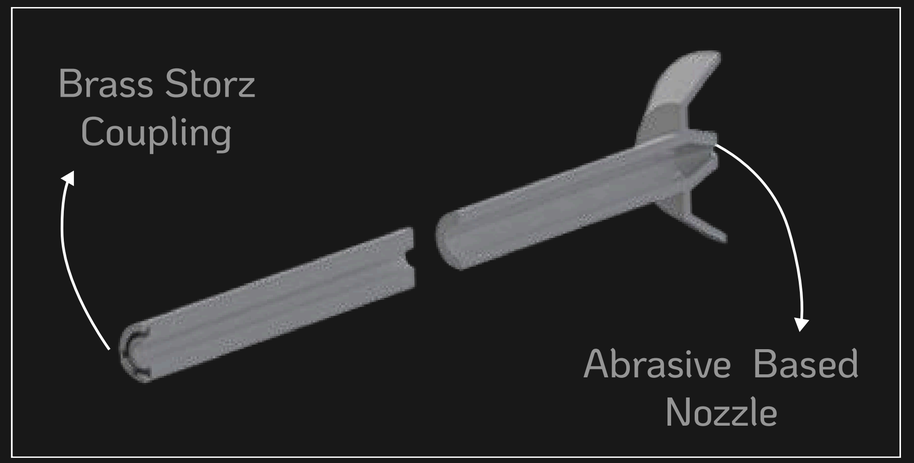


Tool Breakdown

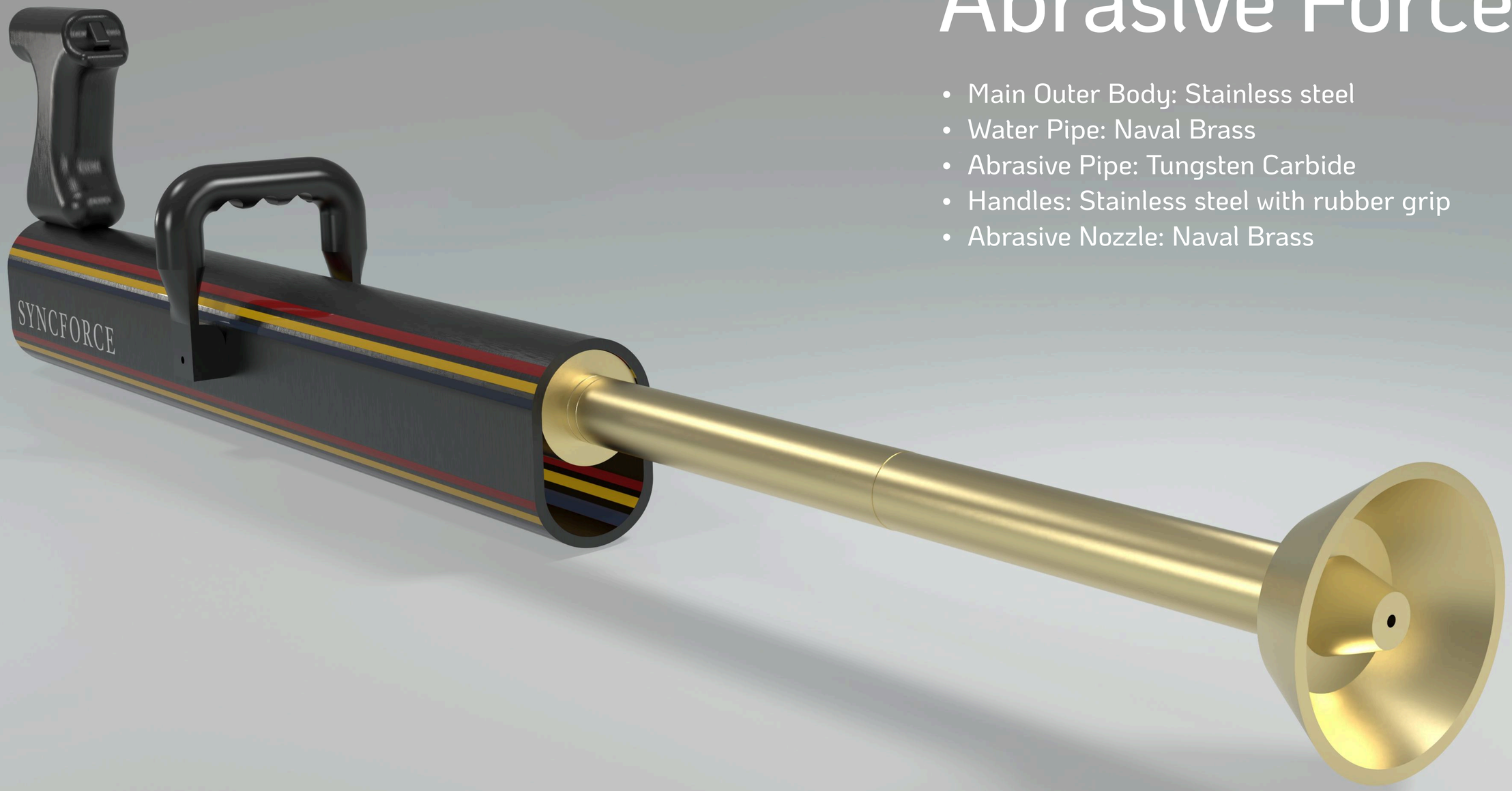








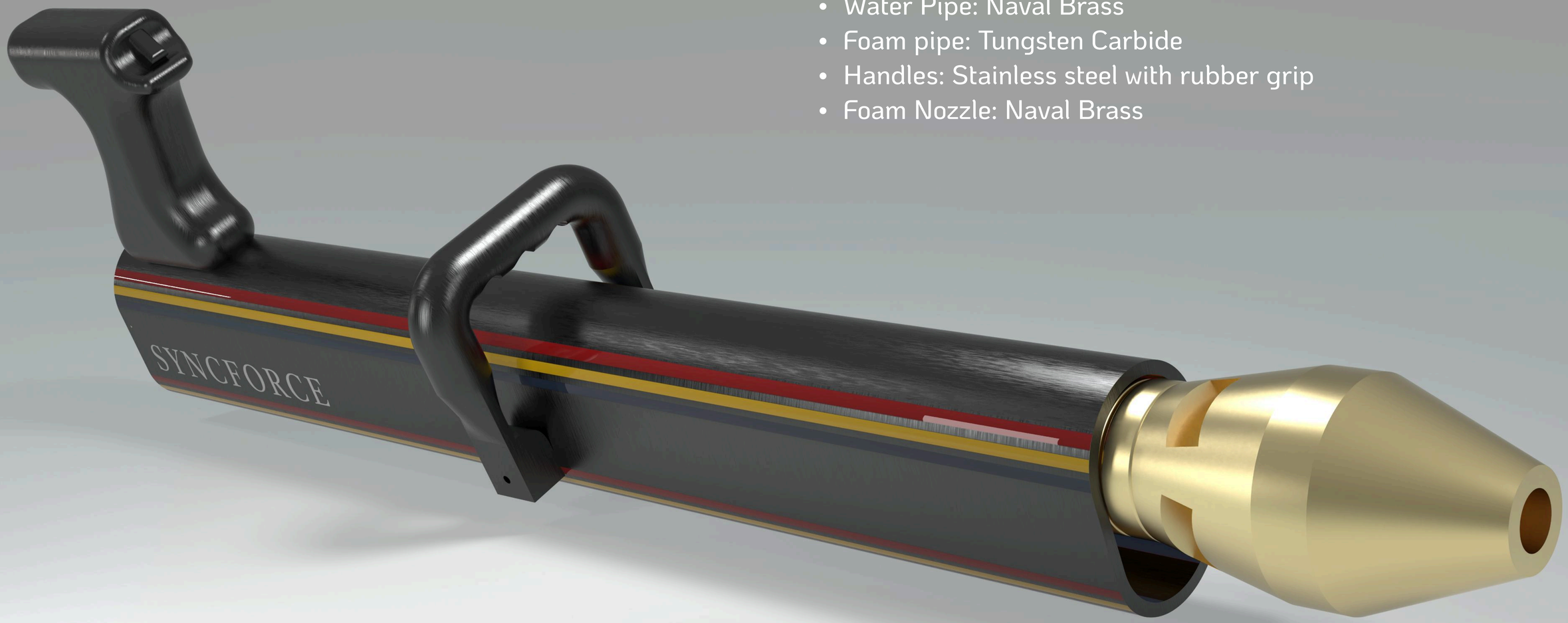
Abrasive Force



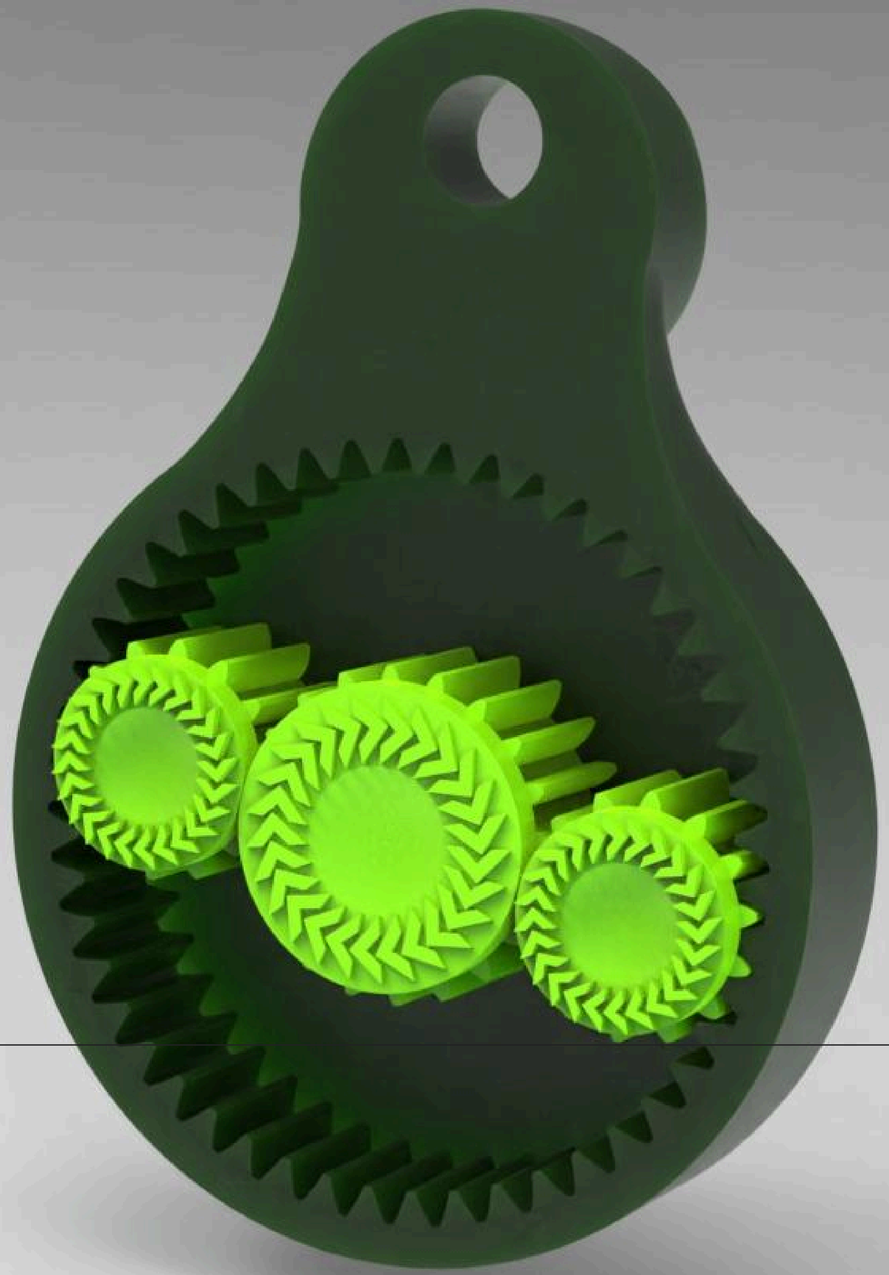
- Main Outer Body: Stainless steel
- Water Pipe: Naval Brass
- Abrasive Pipe: Tungsten Carbide
- Handles: Stainless steel with rubber grip
- Abrasive Nozzle: Naval Brass

Foam Extinguisher

- Main Outer Body: Stainless steel
- Water Pipe: Naval Brass
- Foam pipe: Tungsten Carbide
- Handles: Stainless steel with rubber grip
- Foam Nozzle: Naval Brass



AvoSpin



Project Duration: 2 months
Type: Elective
Team: Individual Project

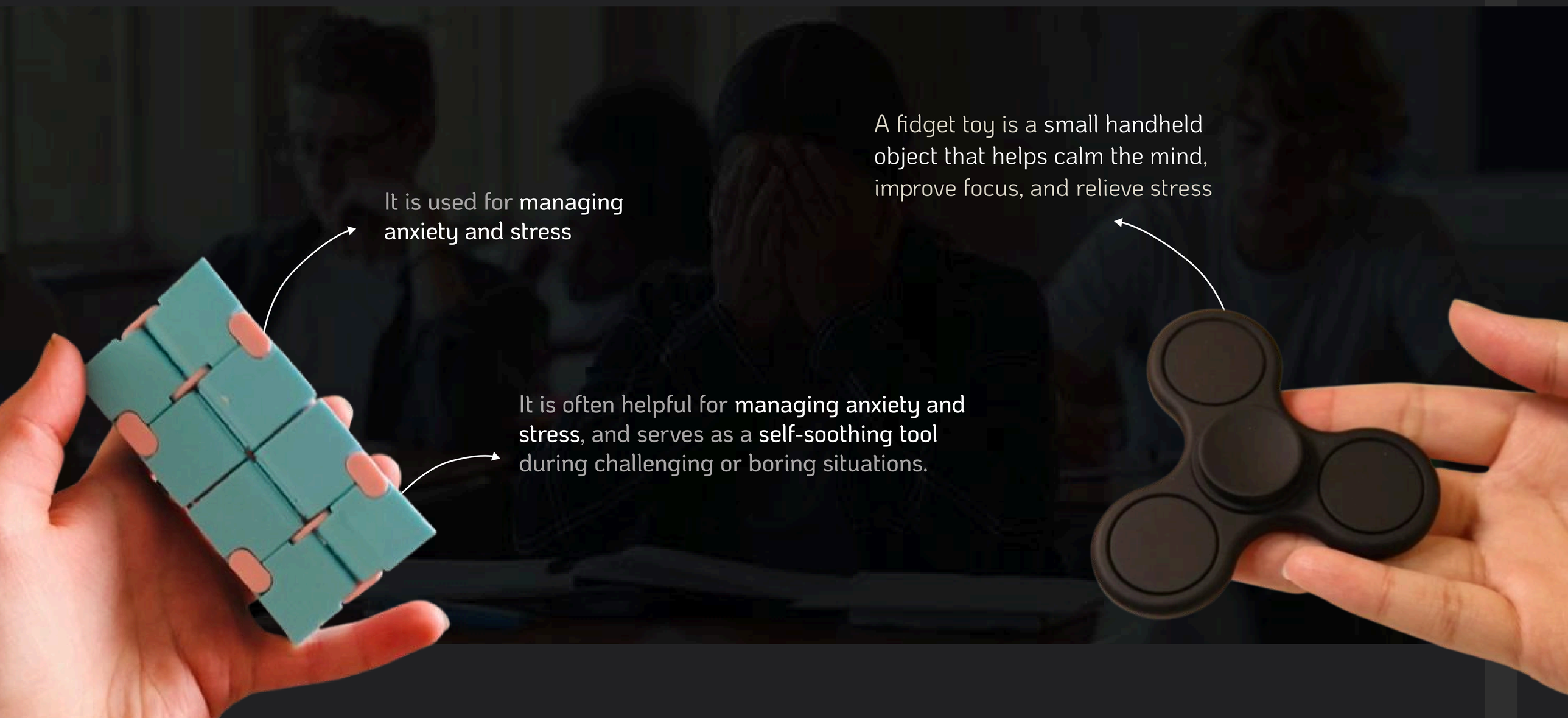
Health Factor

Dealing with mental health as a teen

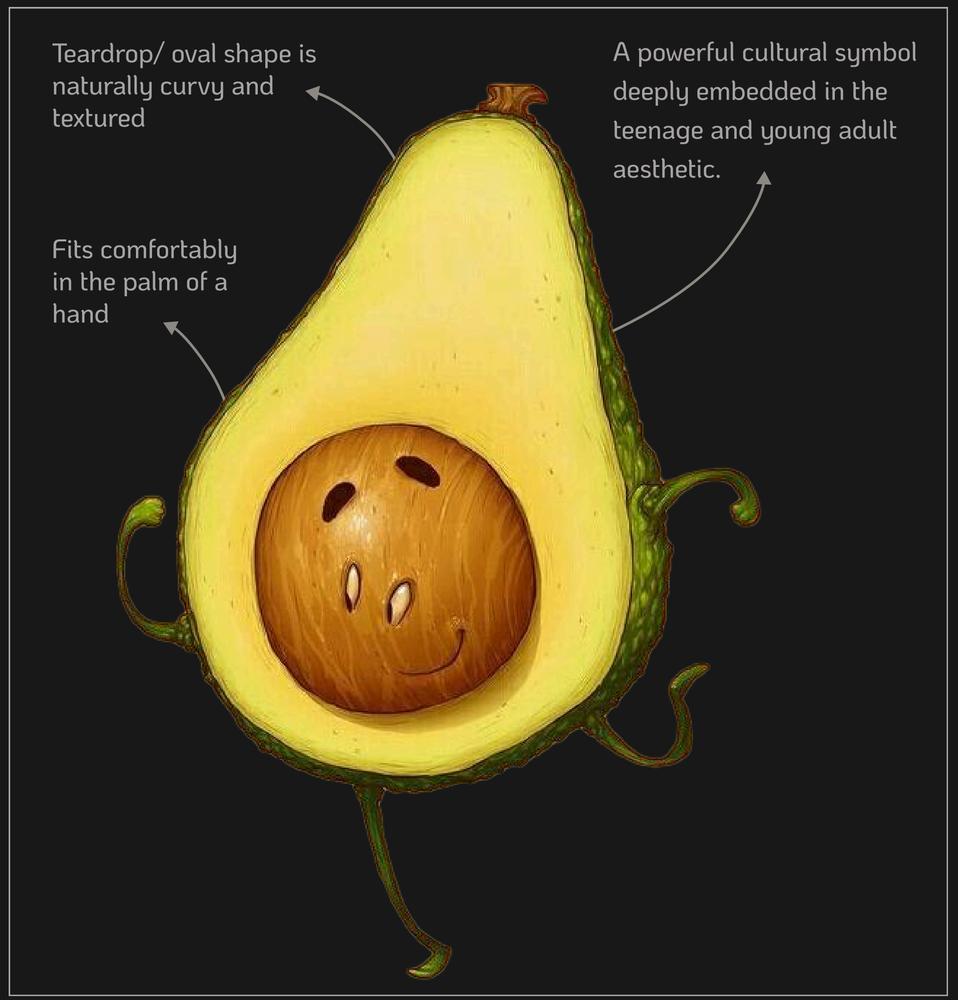
A fidget toy is a small handheld object that helps calm the mind, improve focus, and relieve stress

It is used for managing anxiety and stress

It is often helpful for managing anxiety and stress, and serves as a self-soothing tool during challenging or boring situations.



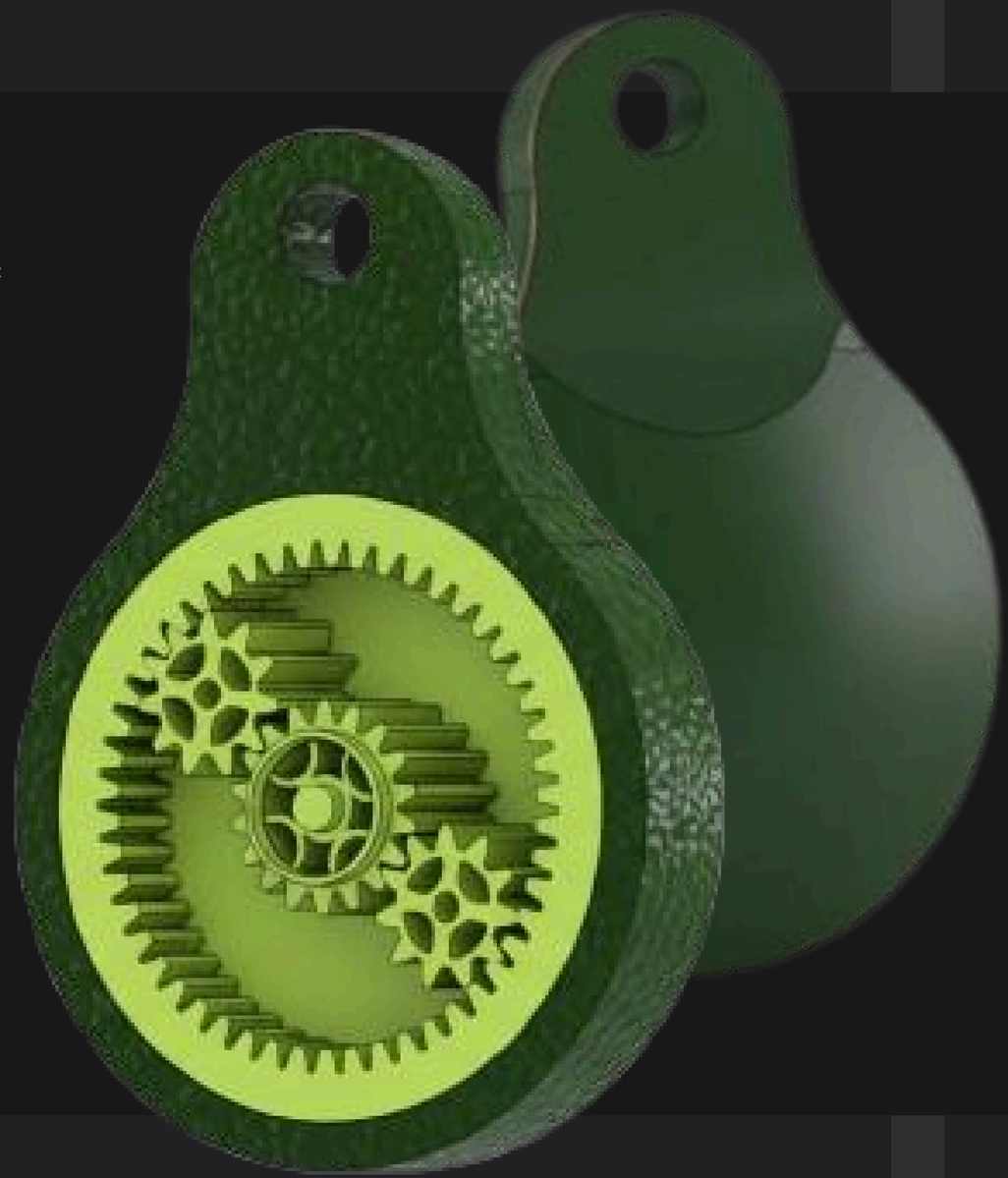
What is AvoSpin?

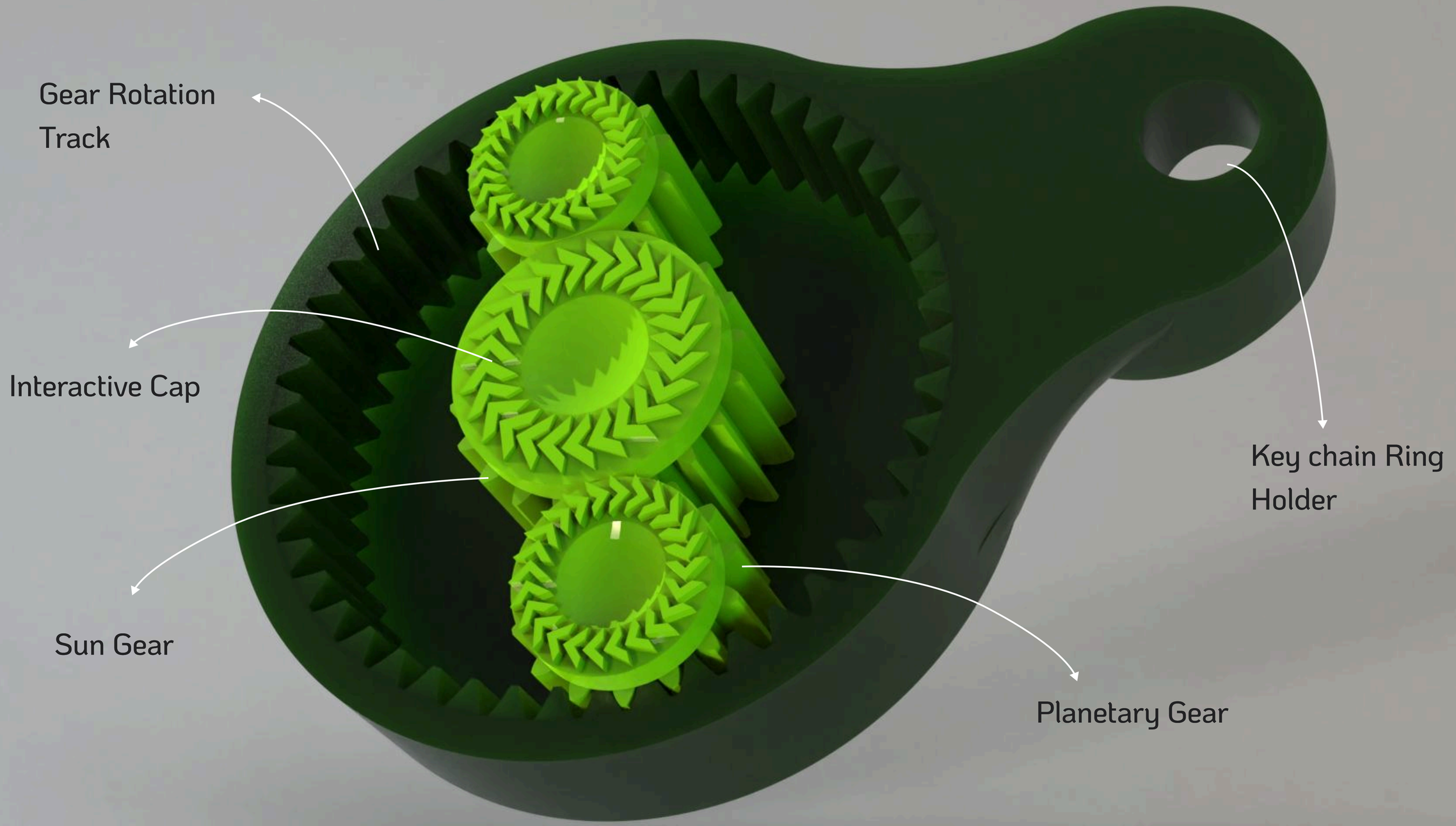


- AvoSpin is a portable sensory accessory designed for teens seeking focus and stress relief
- Functional planetary gear system multiple gears rotate together in a compact configuration, providing satisfying motion and tactile feedback.

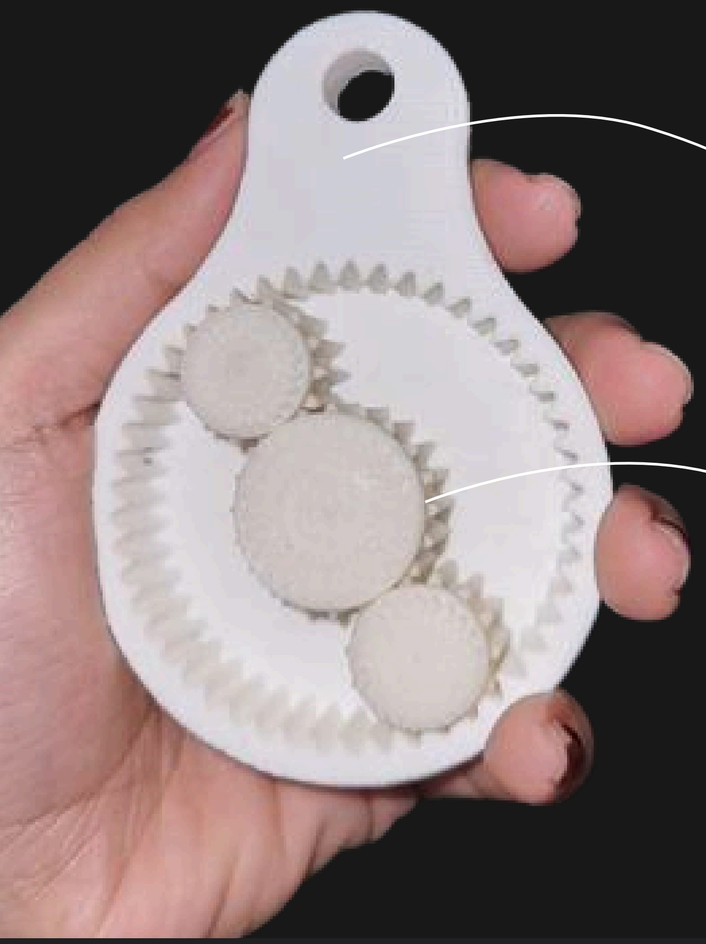
Components:

- Sun gear
- Planetary gear
- Rack
- Main body
- Ball bearings



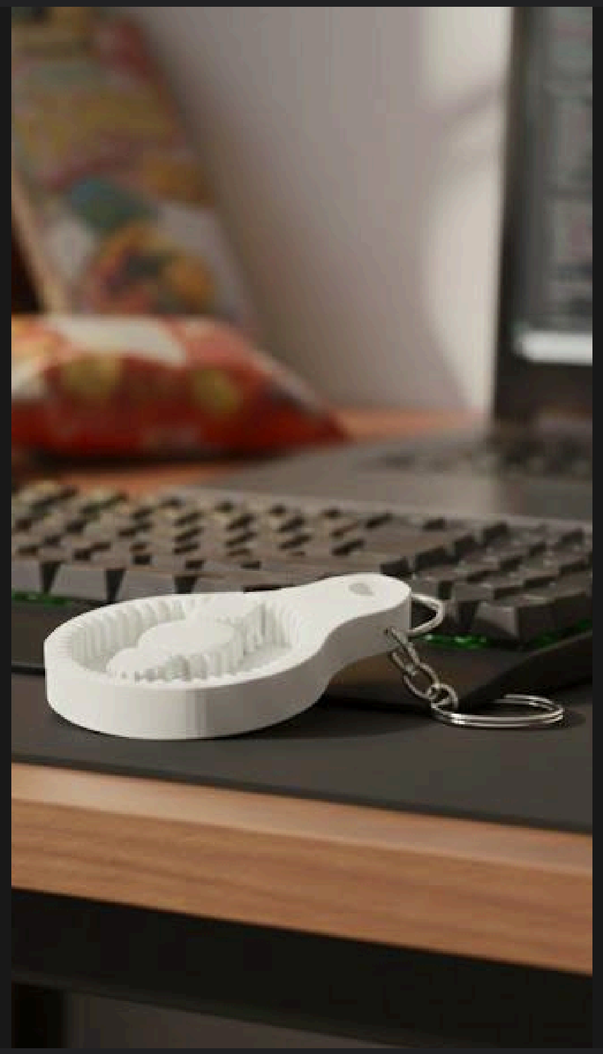

Gear Rotation Track
Interactive Cap
Sun Gear
Key chain Ring Holder
Planetary Gear

Prototyping



Easy to interact and compact design

Smooth gear operation based on ball bearings





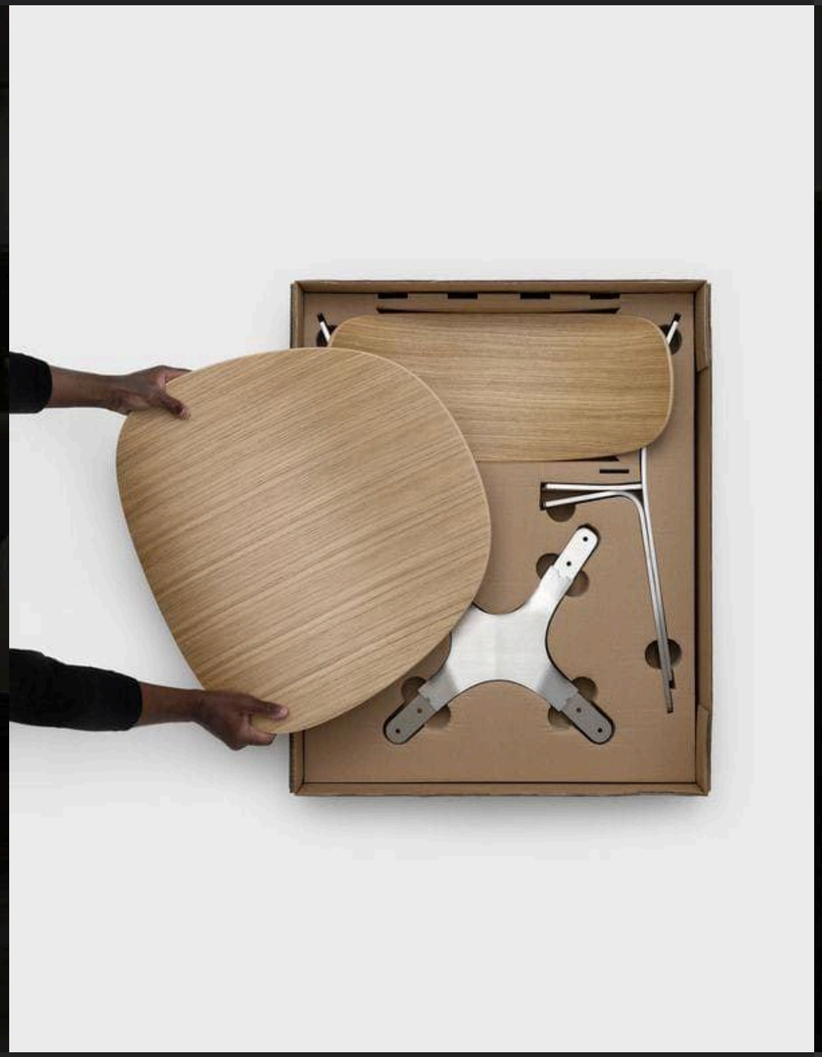
Modulo

Project Duration: 4 months
Type: Product design studio project
Team: Individual Project



Context

To design a household chair that uses a durable joinery system for easy assembly and disassembly without specialized tools. The chair must be optimized for flatpack packaging to minimize shipping volume and storage space. It should combine functionality, structural integrity, and modern aesthetics, making it suitable for efficient mass production and should be appealing for contemporary home environments.

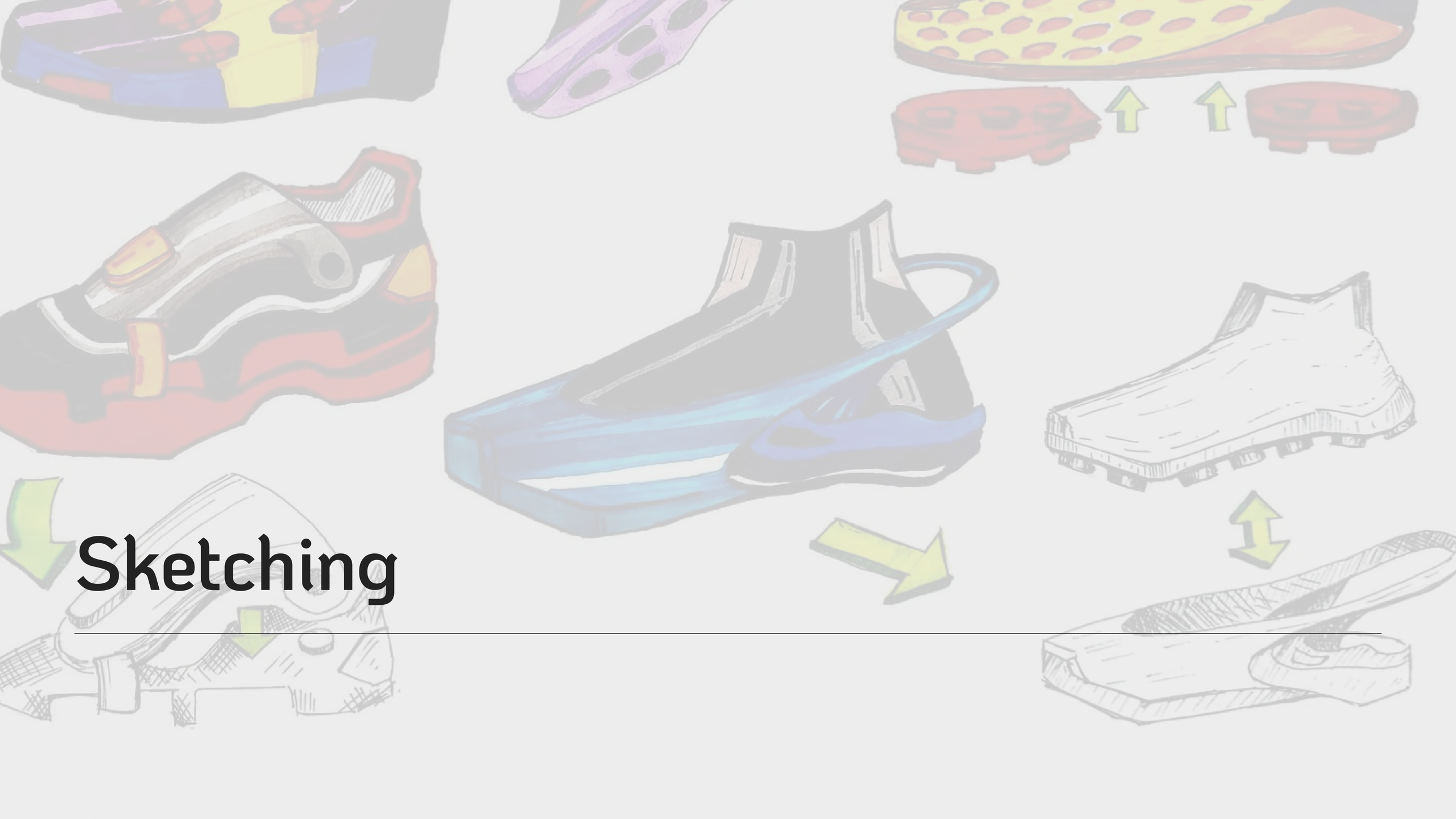


Ideation

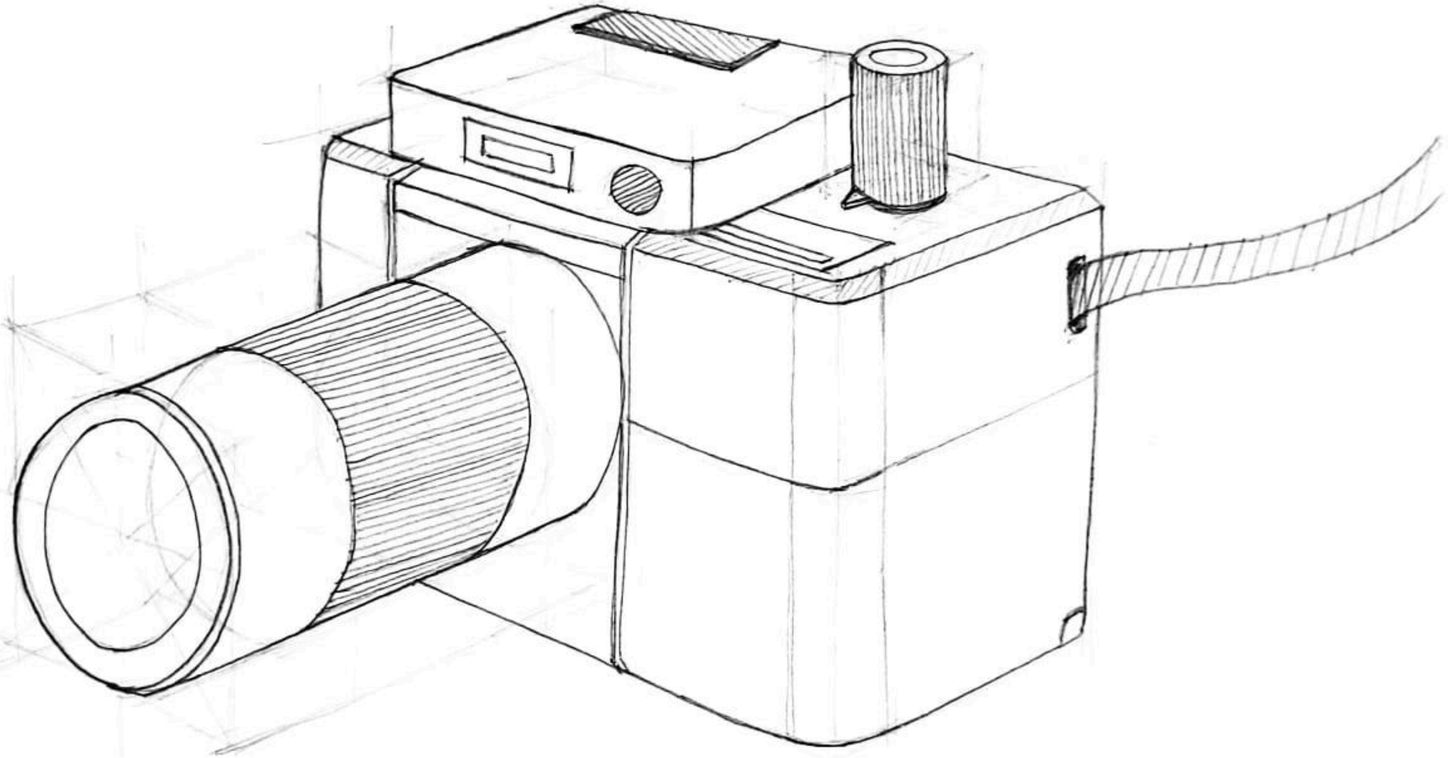

BACK REST
SECOND JOINERY
METAL JOINT
FRONT LEG
BACK
BACK REST
JOINERY GROOVE
JOINERY EXTENSION
GROOVE SLIT

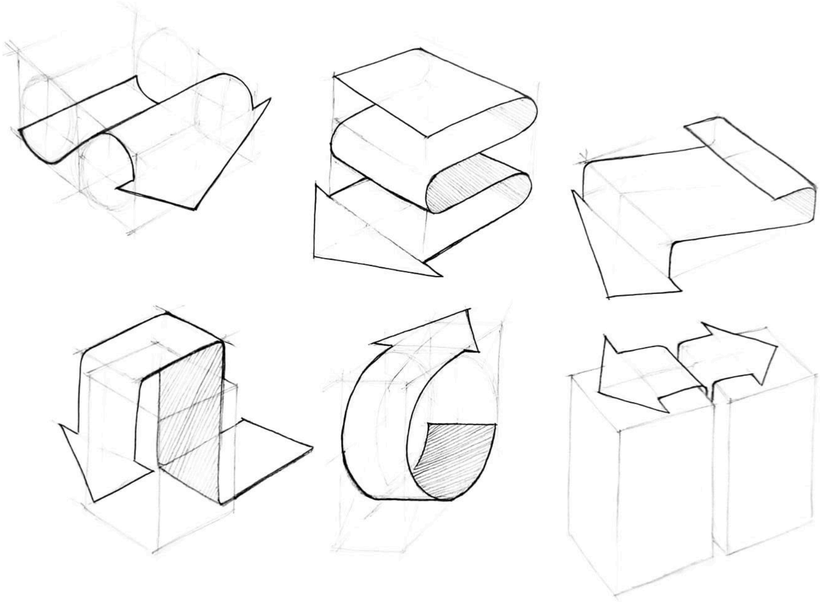


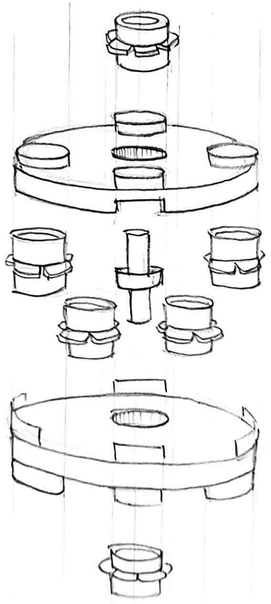
What is Modulo?
A flat pack living room furniture, easy to assemble with only
4 components joined by simple screwing based joinery

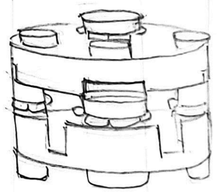


Sketching

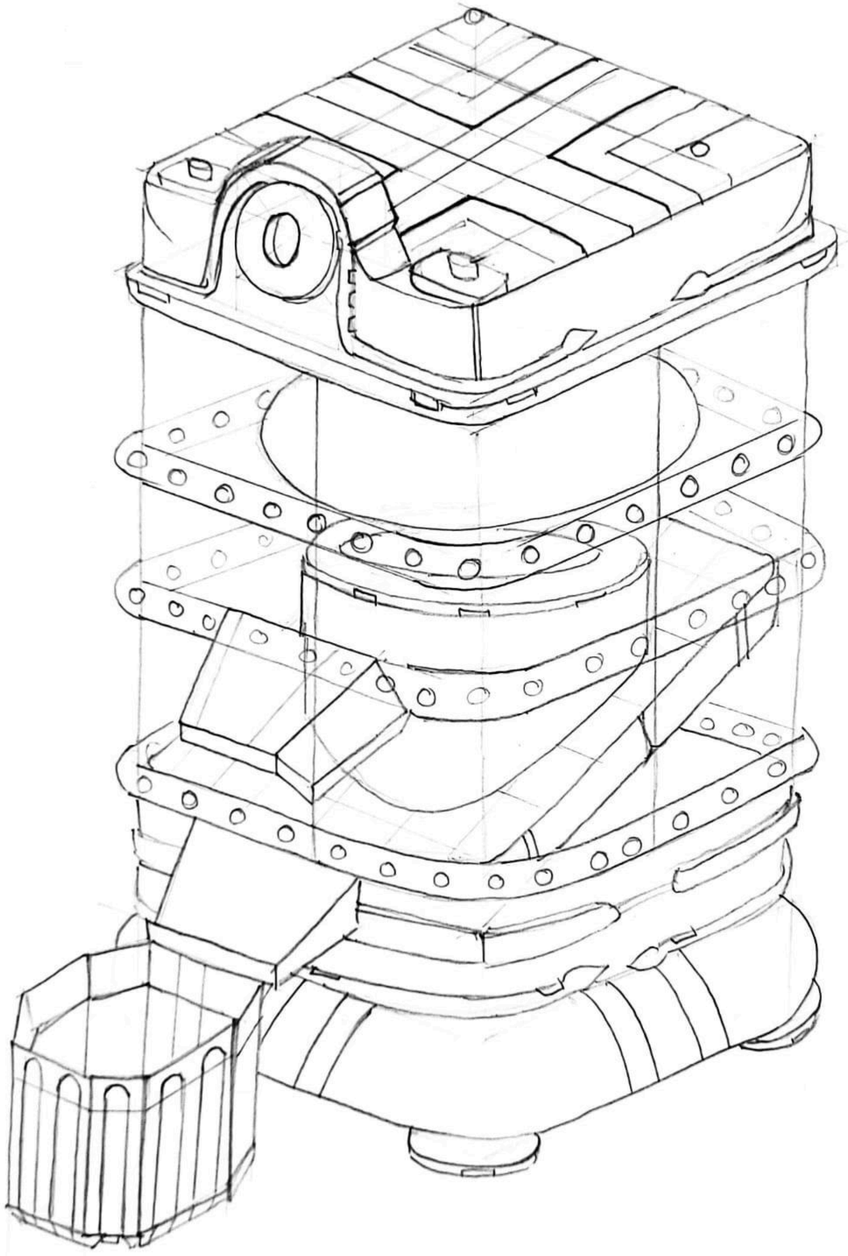


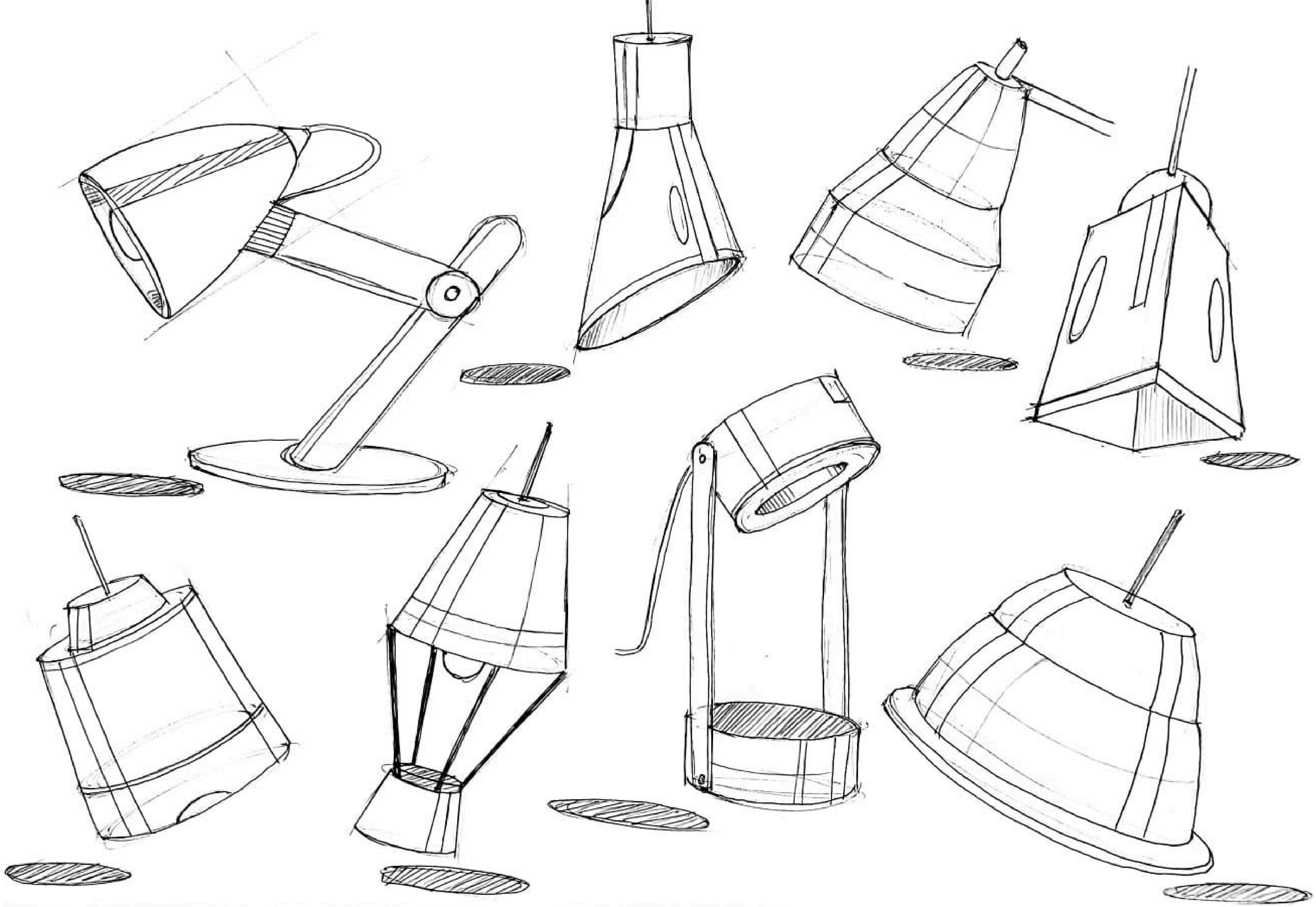


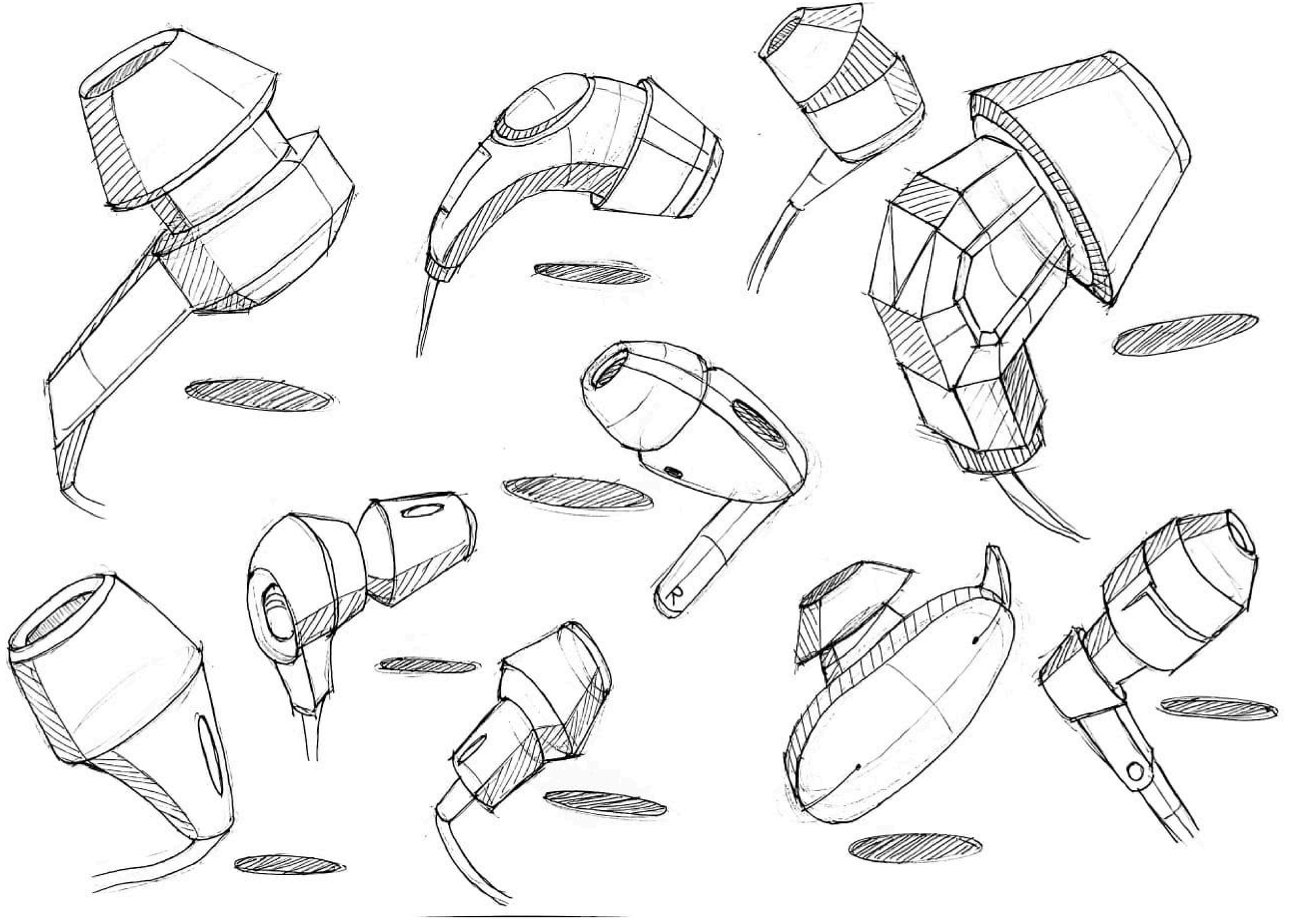




MINIATURE
POPCORN






R
O